

Material Dispersion in Optical Fiber Waveguides

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In weakly guiding dielectric (glass) fiber waveguides the factors governing the envelope delay distortion of optical signals can be separated conveniently into material and waveguide contributions. By making use of the fact that the normal dispersion of the index-of-refraction of glasses accurately fits a single oscillator Sellmeier equation, simple relations of general validity are obtained for the material contribution to the group delay and frequency dependence of the group delay in dielectric fiber waveguides. Using these results expressions are obtained for the material contribution to the information bandwidth of fiber optic waveguides excited by wideband sources (light emitting diodes) and narrow band sources (lasers). It is shown that in single mode guides the bandwidth limitation imposed by material dispersion is negligible, while in multimode guides the difference in group delay of higher order modes imposes the most severe bandwidth limitation.

Introduction

The delay distortion experienced by optical signals in dielectric fiber waveguides is due basically to the effects of material dispersion and to waveguide characteristics. In a recent paper Gloge¹ has shown that for weakly guiding fibers (i.e., fibers having small refractive index differences between core and cladding) material and waveguide dispersion effects can be separated as additive effects. The purpose of this paper is to calculate in a general and simple way the material contribution to the group delay and frequency dependence of the group delay in weakly guiding fibers. By using the results of a previous analysis² for the normal dispersion of the electronic dielectric constant of solids, simple relations of general validity are obtained for the delay distortion due to material effects.

Dispersion of the Optical Dielectric Constant

It has been shown² that the normal dispersion of the electronic (optical) dielectric constant in materials accurately satisfies a Sellmeier relation of the form

$$n^2 - 1 = E_0 E_d / (E_0^2 - E^2), \quad (1)$$

where n is the refractive index, E is the photon energy, and E_0 and E_d are, respectively, oscillator energy and dispersion energy parameters. To a good approximation the oscillator energy E_0 is related to the lowest direct bandgap E_i by $E_0 \approx 1.5 E_i$. In single crystals the dispersion energy E_d is found empirically² to scale with nearest neighbor cation coordination number N_c , anion valency Z_a , and effective number of dispersion

electrons N_e (usually $N_e = 8$), i.e., $E_d = \beta N_c Z_a N_e$. The scaling constant β is two valued taking on discrete ionic, β_i , and covalent, β_c , limits ($\beta_i = 0.26$ eV and $\beta_c = 0.37$ eV). Thus crystals with the same chemistry and structure have the same value of E_d . As a result, in the region of transparency ($E \ll E_0$) we obtain from Eq. (1) the relation,

$$n^2 - 1 = E_d / E_0, \quad (2)$$

indicating that to a good approximation the dielectric susceptibility varies inversely with E_0 or the bandgap.

Because glasses are amorphous materials, one might expect E_d to be smaller than in single crystals due to broken bonds, i.e., a smaller N_c . This is certainly borne out when one compares E_d for crystalline quartz (SiO_2) $E_d = 18.3$ eV with E_d for fused silica (SiO_2) $E_d = 14.7$ eV. The oscillator energy for both crystal quartz and fused silica is $E_0 \approx 13.5$ eV. When we compare the oscillator and dispersion energies for a large group of representative glasses we find no systematic trends in E_d as observed in single crystals.

Shown in Fig. 1 are plots of $1/(n^2 - 1)$ vs E^2 (or $1/\lambda^2$) for several glasses. These glasses were selected from the Schott catalog (except SiO_2) and span the range of indices of refraction available from typical crown and flint glasses ($1.45 < n < 2$). The intercept at $E = 0$ in Fig. 1 gives E_0/E_d , and the slope gives $1/E_0 E_d$. Note that the index data fit the single oscillator Sellmeier expression very well. A large number of the glasses given in the Schott catalog have been analyzed in terms of Eq. (1). Table I lists values of E_0 and E_d for a representative sample of glasses together with the code designation given in the Schott catalog. The Sellmeier oscillator parameters listed in Table I should be compared with the extensive listing of parameters for single crystals given in Ref. 2.

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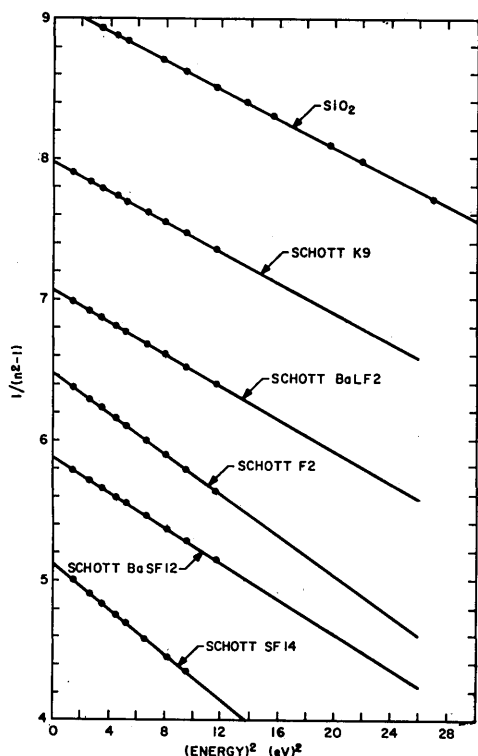


Fig. 1. Plot of refractive-index factor $1/(n^2 - 1)$ vs E^2 for several selected glasses.

Group Delay and Frequency Dependence of Group Delay-Material Contribution to Information Bandwidth of Fibers

The group velocity in a dispersive medium is given by

$$v_g = c/(n + E dn/dE), \quad (3)$$

where E is the photon energy and c is the speed of light. In the region of transparency Eq. (1) accurately represents the dispersion of the optical susceptibility. By differentiating Eq. (1) we find that to an excellent approximation in the region $E \ll E_0$ (i.e., in the visible for glass materials)

$$dn/dE = E_d E / n E_0^3. \quad (4)$$

Thus the group delay per unit length $T = 1/v_g$ is given by

$$T = (n/c)[1 + (E_d E^2 / n^2 E_0^3)]. \quad (5)$$

Using the oscillator parameters listed in Table I we conclude that throughout the visible part of the spectrum the material contribution to the group delay is simply n/c , the value in a nondispersive medium.

Also of interest is the frequency dependence of the group delay. In optical bandwidth B (expressed in units of energy) the differential group delay per unit length is given by

$$s = B(d/dE)(1/v_g). \quad (6)$$

It can be shown from Eqs. (1) and (3), after some algebra, that in the region of transparency $E \ll E_0$,

$$s \approx 3BE_d E / cnE_0^3. \quad (7)$$

This is the material contribution to the waveguide time dispersion. Gloge¹ has shown that the total waveguide time dispersion is the superposition of the material contribution and a negative waveguide characteristic. Using values appropriate to SiO_2 we find that in the visible $E \approx 2 \text{ eV}$,

$$cs/\Delta \approx 0.05, \quad (8)$$

where $\Delta = B/E$ is the relative bandwidth. Referring to the data in Table I we conclude that for most glasses E_0 is smaller than the 13.4 eV value for SiO_2 , and, as a consequence, the cs/Δ values will be somewhat higher than indicated by Eq. (8).

If a Gaussian pulse of width τ is transmitted through a dispersive medium of length L the output pulse width τ_0 is

$$\tau_0^2 = \tau^2 + s^2 L^2. \quad (9)$$

This equation applies for a pulse whose spectral width is small compared to the emission spectrum of the source¹ or for a bandwidth-limited Gaussian pulse.³ In the first case B in Eq. (7) is the spectral width of the source, while in the second case B is the reciprocal of the pulse width, i.e., $B = h/\tau$, where h is Planck's constant ($h = 4.14 \times 10^{-15} \text{ eV}\cdot\text{sec}$). Thus for a broadband source (e.g., an electroluminescent diode) transmitting a wide pulse we have

$$s \approx 3\Delta E_d E^2 / cnE_0^3 \approx 170\Delta(\text{nsec/km}), \quad (10)$$

whereas for a bandwidth-limited pulse from, for example, a mode-locked laser we have

Table I. Dispersion Parameters for a Representative Selection of Glasses from the Schott Catalog

Material	n at $\lambda = 0.588 \mu$	E_0 (eV)	E_d (eV)
SiO_2	1.4586	13.4	14.7
PK2	1.5182	12.96	15.9
PSK2	1.5687	12.51	16.94
BK1	1.5101	12.64	15.72
K9	1.5148	12.2	15.3
BaLF2	1.5710	11.35	16
BaK4	1.5688	11.86	16.8
SK20	1.5596	12.42	17.3
KF6	1.5174	11.5	14.5
BaLF2	1.5710	11.25	15.94
SSK4	1.6177	11.67	18.25
LLF2	1.5407	10.86	14.36
BaF9	1.6433	10.8	17.67
LF1	1.5731	10.26	14.5
F2	1.6200	9.4	14.5
F6	1.6364	9.3	14.8
BaSF12	1.6700	9.74	16.6
SF6	1.8052	7.78	16.23
SF14	1.7618	8.1	15.9
LaK N14	1.6968	11.7	21.2
LaF N2	1.7440	10.37	20.3
LaSF N3	1.8061	9.13	19.6

$$s \approx \frac{3\hbar E_d E}{cnE_0^3 \tau} \approx \frac{0.35}{\tau} \times 10^{-12} \frac{\text{nsec}}{\text{km}}, \quad (11)$$

where τ is the pulse width expressed in seconds. Using Eqs. (10) and (11) we can calculate a material contribution to the bandwidth of a dielectric fiber waveguide. When $sL = \tau$ the output pulse width is $\sqrt{2}\tau$. We define the base bandwidth or information bandwidth of the fiber as $b = 1/\tau = 1/sL$. In this base bandwidth negligible envelope delay distortion occurs due to the frequency dependence of the group delay. Accordingly for a broadband source we have from Eq. (10),

$$bL_{\text{km}} \approx cnE_0^3/3\Delta E_d E^2 \approx (5.9/\Delta)\text{MHz-km}, \quad (12)$$

or

$$\tau \approx \frac{3\Delta E_d E^2}{cnE_0^3} L \approx 170\Delta L_{\text{km}} \text{ nsec}, \quad (13)$$

where L_{km} is the length expressed in kilometers. For a bandwidth-limited pulse Eq. (11) gives

$$b\sqrt{L_{\text{km}}} \approx (cnE_0^3/3\hbar E_d E)^{1/2} \approx 53 \text{ GHz-km}^{1/2}, \quad (14)$$

or

$$\tau \approx [(3\hbar E_d E/cnE_0^3)L]^{1/2} \approx 18\sqrt{L_{\text{km}}} \text{ psec}. \quad (15)$$

We emphasize that Eqs. (10)–(15) only give the material dispersion contribution to the time dispersion of a dielectric fiber waveguide. To calculate the total time dispersion one must add to s the contribution due to waveguide dispersion as discussed by Gloge.¹ We also emphasize that the numerical examples apply specifically to SiO_2 at $E \approx 2 \text{ eV}$. Other glasses will have similar behavior as indicated by the data in Table I.

The analysis leading to Eqs. (12)–(15) applies to each individual waveguide mode in a dielectric fiber waveguide. Gloge¹ has shown that in weakly guiding fibers, where the refractive index difference between core and cladding is small, the total differential group delay s in single mode, and to a good approximation in multimode fibers, is determined principally by material dispersion. Thus in a single mode waveguide the envelope of a pulse will experience a group delay T and an envelope delay distortion determined by the differential group delay s given by Eqs. (10) or (11). Equation (12) implies for a single mode fiber, a bandwidth length limitation of $\approx 600 \text{ MHz-km}$ for a source with 1% relative bandwidth (e.g., a light emitting diode). Equation (14) implies a bandwidth square-root-of-length limitation of $\approx 50 \text{ GHz-km}^{1/2}$ for a bandwidth limited laser source. In multimode fibers each mode experiences a different group delay T_m but, to a good approximation, the same differential group delay s . Therefore, in a multimode waveguide the envelope delay distortion of a pulse is determined by the difference in arrival times or group delay of different higher order modes, as well as by the differential group delay s . It can easily be shown that the group delay of higher order modes imposes a more serious bandwidth limitation than the material dis-

persion contribution. For example, for a multimode fiber having a core-cladding index difference of 10% the bandwidth can be as low as 1 MHz-km.

We may conclude that the bandwidth limitation due to material dispersion in single mode guides is negligible for most glasses of interest, and that the bandwidth limitation due to the difference in group delay of higher order modes is of primary importance in multimode guides with index differences larger than 1%. In a recent study Kapron and Keck⁴ have numerically calculated the information bandwidth of a single mode fiber including both material and waveguide dispersion effects. Their numerical calculations and our bandwidth estimates for single mode fibers are in good agreement.

Conclusions

We have shown that the dispersion of the index of refraction of glasses accurately fits a single oscillator Sellmeier equation. Two parameters, an oscillator energy E_0 and a dispersion energy E_d , completely characterize the refractive properties of glasses in the visible portion of the spectrum. This result has been reported previously for single crystals,² where it was shown empirically that the dispersion energy is a unique parameter related to crystal structure, chemistry, and ionicity. No simple behavior has been found for the dispersion energy parameter of glasses.

Gloge¹ has shown previously that in weakly guiding fibers, material and waveguide contributions to the group delay and differential group delay can be separated. By combining the results of his analysis together with the Sellmeier description of the index of refraction, simple equations of general validity have been obtained for the material contribution to the group delay and frequency dependence of the group delay in the visible region of the spectrum. In dielectric fiber waveguides the material contribution is the dominant contribution to the frequency dependence of the group delay. Our results indicate that the limitation to the information bandwidth imposed by the material is negligible in single mode and multimode dielectric fiber waveguides. In multimode guides the difference in group delay experienced by higher order modes imposes the most severe bandwidth limitation.

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